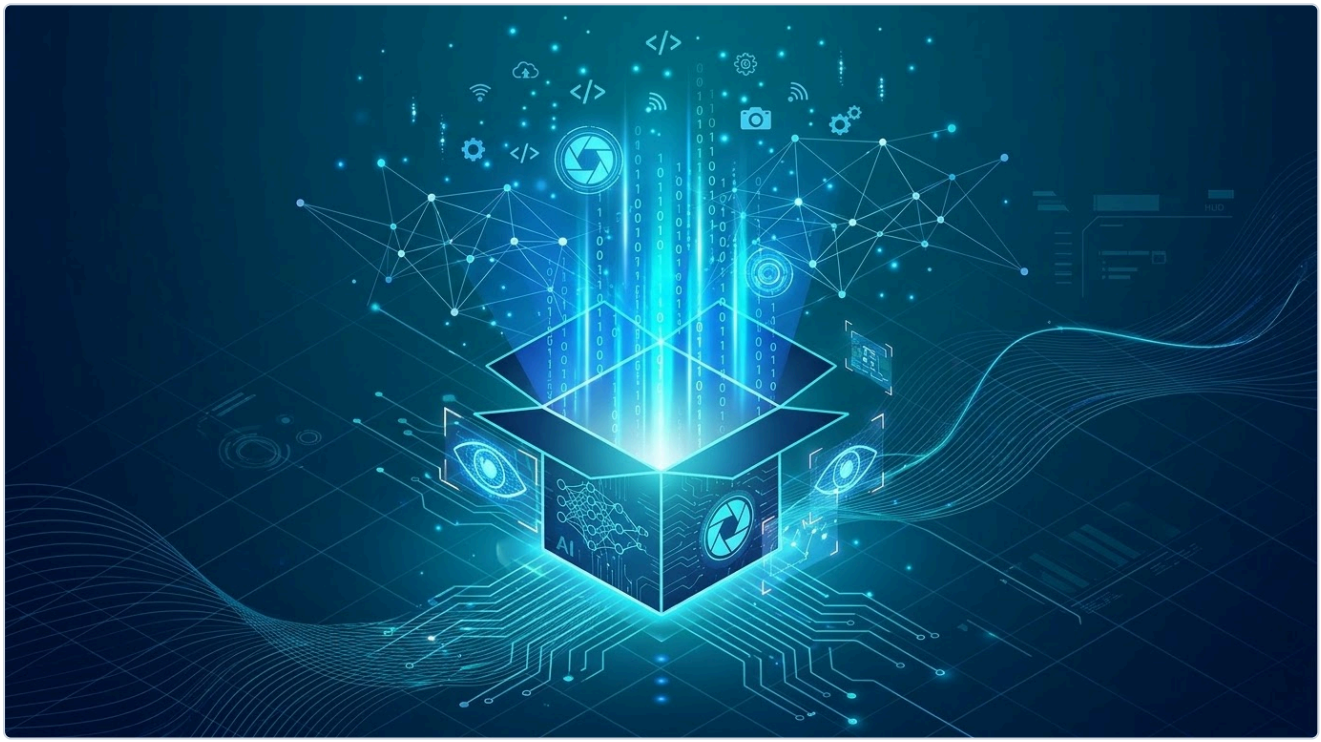
# OpenAA: Architecture & Ecosystem Overview

**OpenAA (Open Application Architecture) is a large-scale, open-source C++ software framework dedicated to high-end industrial visual inspection systems. Developed by Megatrade Corporation, OpenAA was created to break away from traditional "black-box" proprietary systems, democratize machine vision, and foster a global, collaborative engineering ecosystem.**



*Democratizing machine vision through an open and collaborative platform.*

## 1. Background & The Challenge

In modern manufacturing, almost every production process requires meticulous visual inspection to ensure quality. However, the visual inspection equipment market has traditionally been fragmented and inefficient.

Currently, companies face significant challenges:

- **The High Cost of Customization:** Every production process is unique, requiring custom-built inspection equipment. Proprietary systems often cost millions of yen, placing a heavy burden on smaller factories.
- **The "Black Box" Problem:** Most commercial inspection systems are closed ecosystems. Users cannot access the source code or modify the underlying algorithms, leading to vendor lock-in.



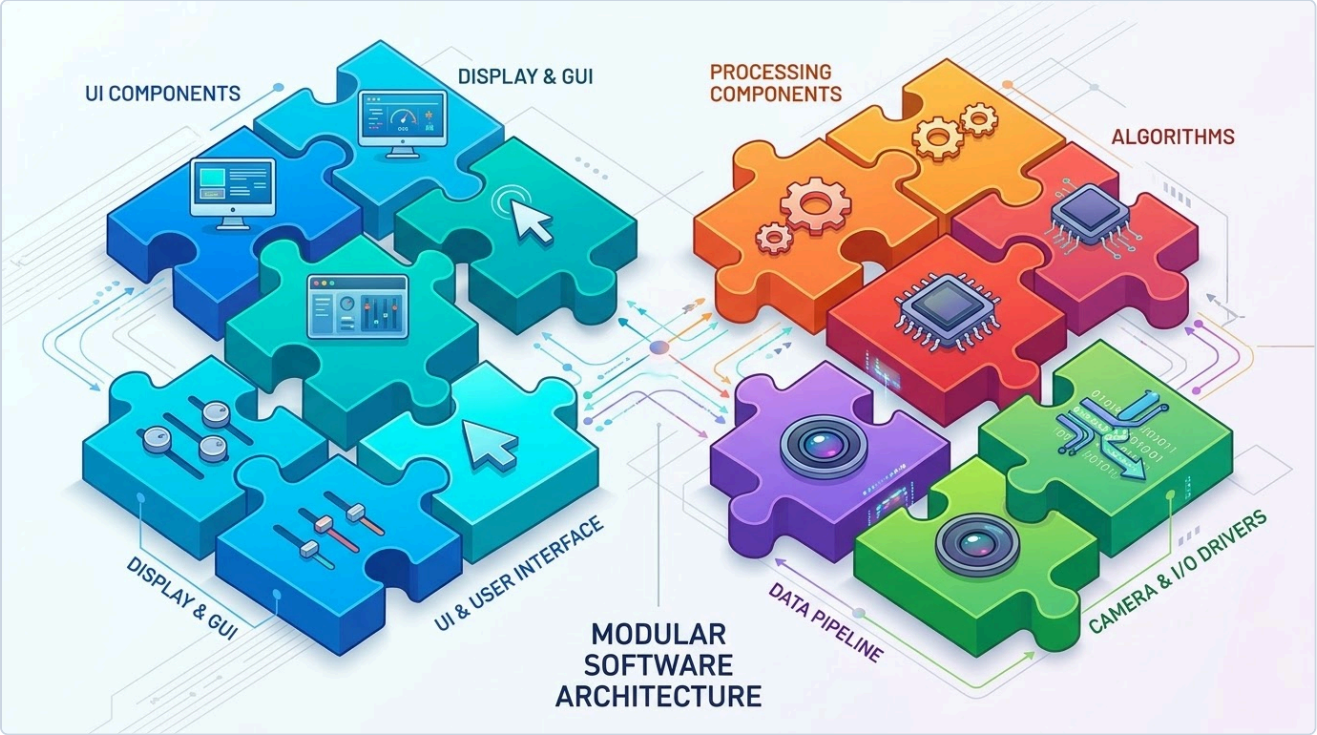
*Complex modern industrial manufacturing lines demand flexible and cost-effective inspection solutions.*

OpenAA was created to solve these problems by providing a completely free, open-source C++ framework for industrial visual inspection.

## 2. OpenAA Software Architecture

### Componentization & UI/Processing Separation

OpenAA is built around a highly modular, component-based architecture consisting of over 1,200 independent components (DLL/so files).




*Strict separation between UI components and processing algorithms.*

Key architectural benefits:

- **Strict Separation of UI and Processing:** The user interface components and the core processing algorithms are completely separated and communicate via internal message packets.
- **Team Development & Scalability:** Because components are independent, multiple developers can work simultaneously without conflicts. If a specific algorithm needs an update, only that component needs to be replaced.
- **Cross-Platform:** Built on the Qt framework, OpenAA runs seamlessly on Windows, Linux, and other operating systems.

## Hierarchical Data Structure

All inspection algorithms and data configurations in OpenAA follow a strict hierarchical structure, ensuring that systems can scale from simple setups to massive inspection lines without fundamentally changing the code.

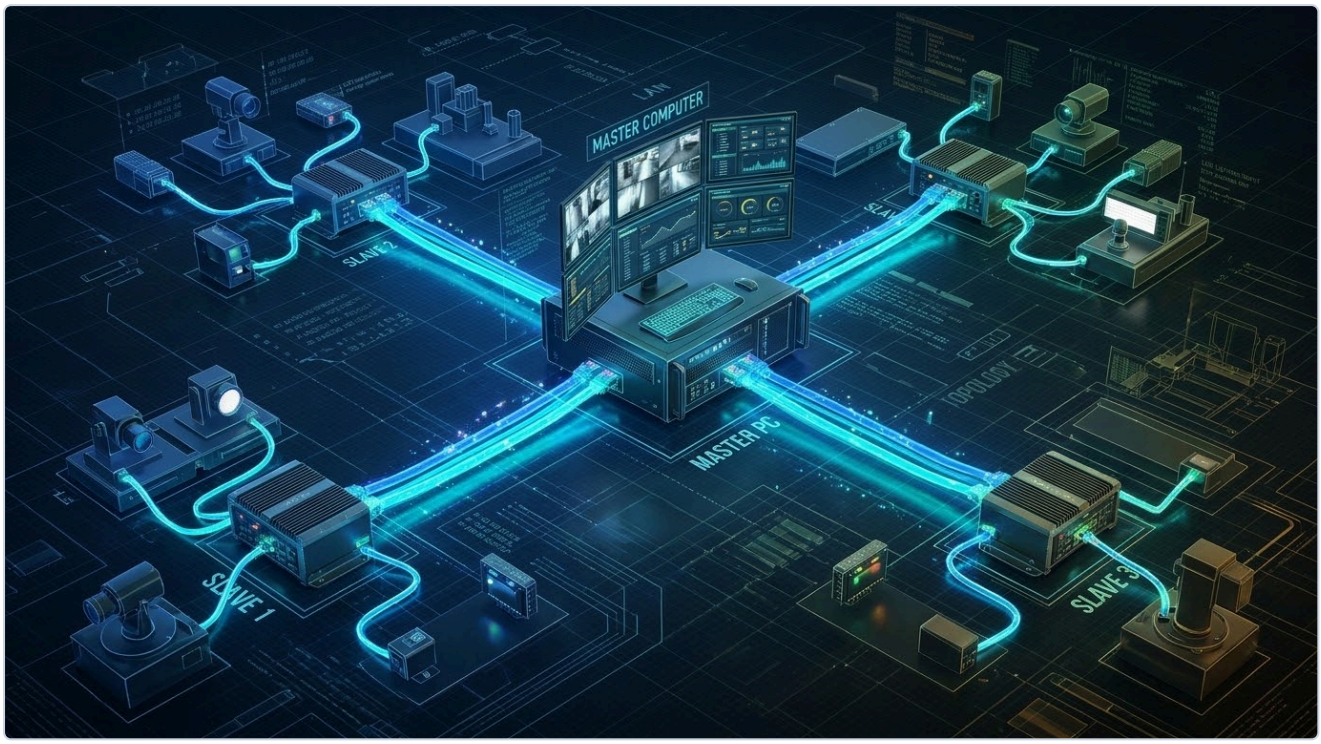



*The hierarchical data structure: Phase, Page, Layer, and Item.*

- **Phase:** Represents multiple inspection passes.
- **Page:** Represents individual cameras or PCs.
- **Layer:** Represents specific attributes (e.g., color, lighting type).
- **Item:** Represents the specific inspection elements and algorithms per region.

## Scalable Network Architecture

Because of the strict separation of UI and processing via message packets, OpenAA inherently supports complex Master-Slave network configurations.




*Master-Slave network configuration for distributed processing.*

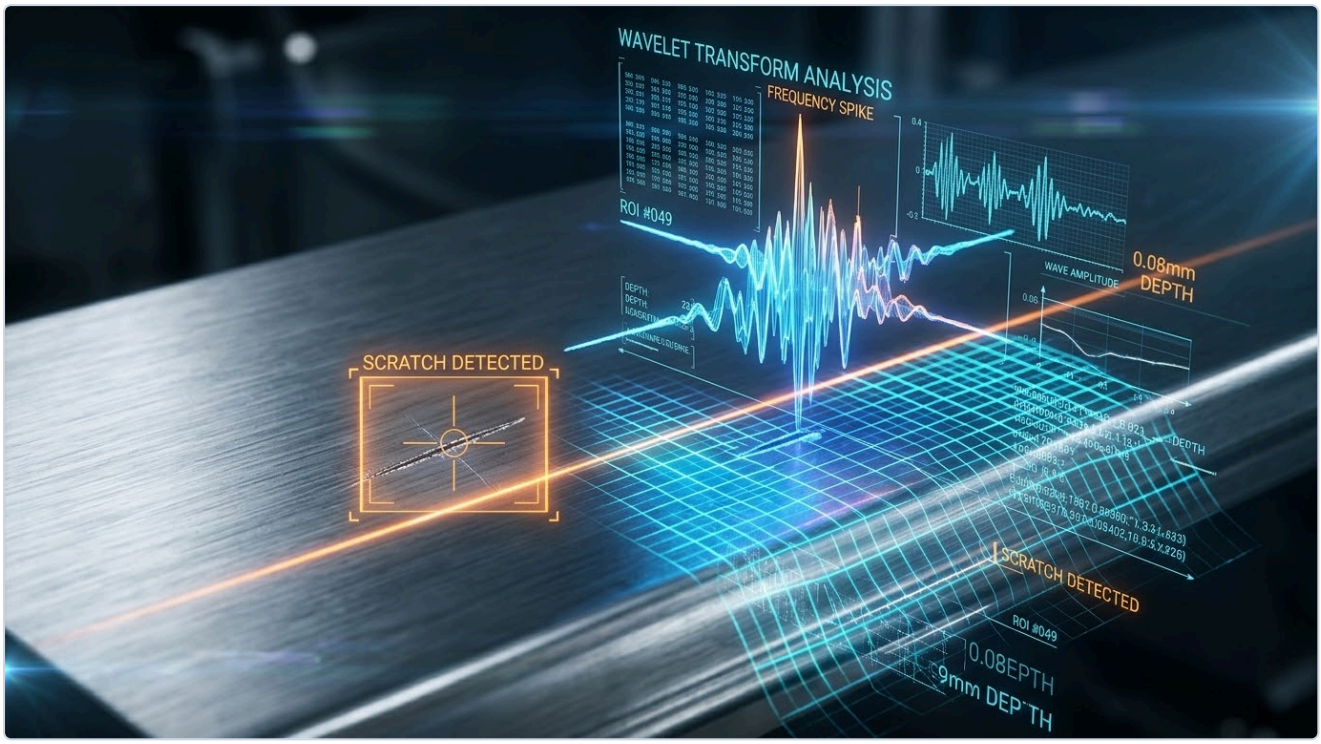
A single software build can function as both the controller (Master) and the execution unit (Slave). The UI on the Master PC can control and display results as if all data were processed locally, while the actual heavy lifting is distributed across multiple Slave PCs.

## 3. Advanced Inspection Capabilities

OpenAA supports state-of-the-art inspection techniques out of the box.

### Defect Enhancement (Wavelet & Projection)

For highly subtle defects, OpenAA utilizes advanced filtering and mathematics.

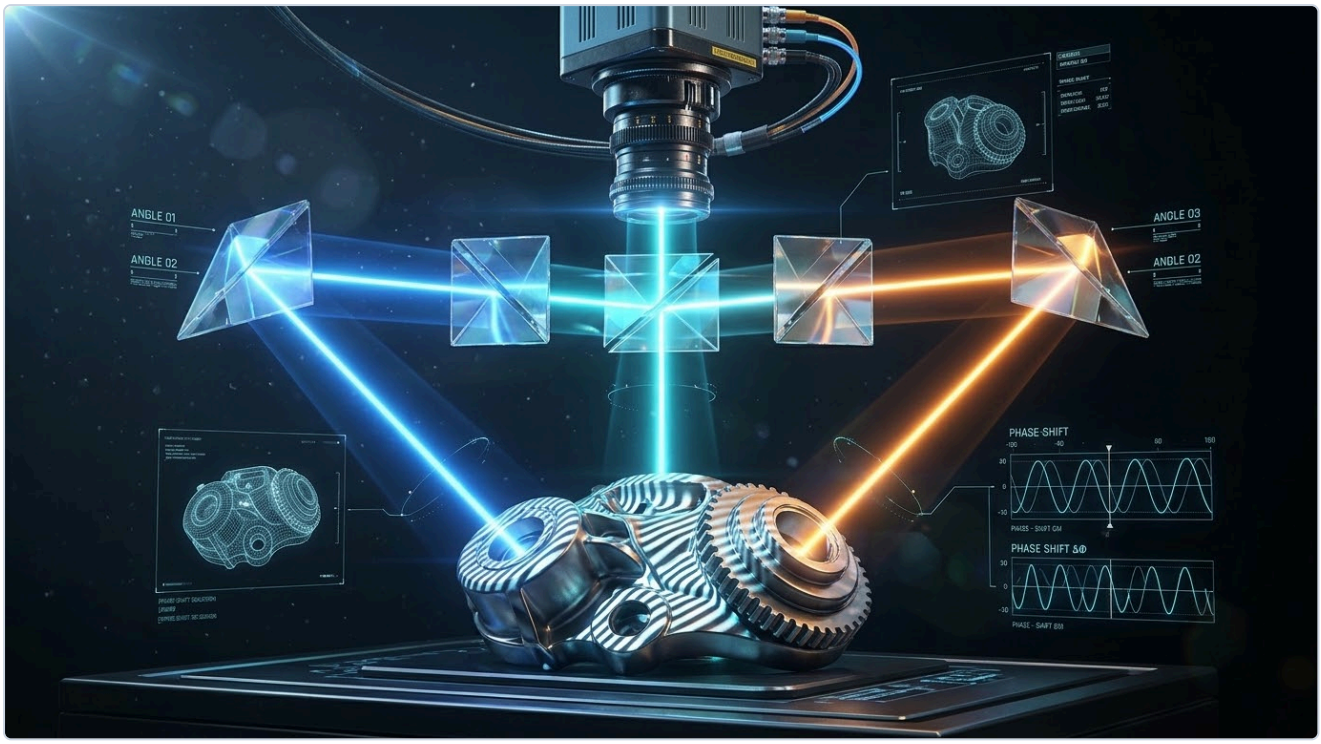



*Enhancing nearly invisible scratches utilizing Wavelet transform analysis.*

By applying Wavelet kernel functions and performing convolution calculations, OpenAA can extract and enhance specific wave-like patterns (such as faint scratches on metal) that are nearly invisible to the naked eye.

## 3D Shape Inspection using Prism Mirrors

By utilizing specialized optical setups, OpenAA can perform 3D shape analysis using a single camera.

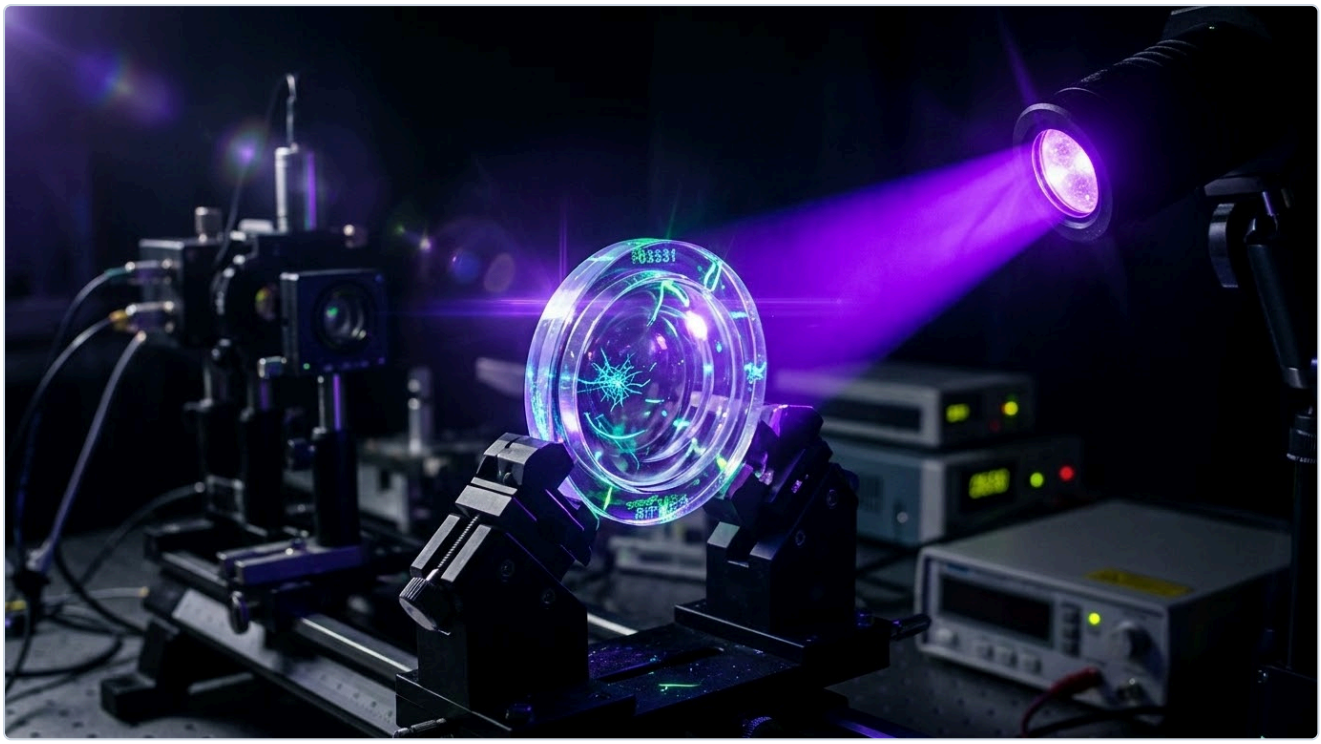



*Capturing 3D topography and phase shifts using prism mirrors.*

By placing prism mirrors around a target, the camera captures three different angles simultaneously. By projecting a fringe pattern (structured light) and measuring the phase shift in the captured lines, OpenAA calculates depth and detects 3D deformation.

## Transparent Object & UV Inspection

Inspecting transparent glass or plastics requires specialized lighting control.



*Revealing hidden defects on transparent surfaces with specialized UV lighting.*

OpenAA seamlessly integrates with various lighting controllers. By switching to specific wavelengths, such as UV light, OpenAA can reveal hidden fluorescent marks, adhesives, or defects on transparent surfaces that standard lighting cannot capture.

## 4. The OpenAA Satellite Ecosystem

To promote the global adoption of OpenAA and solve the business challenges inherent in the inspection equipment industry, we are establishing the **Satellite Company Partnership**.




*A global network connecting OpenAA HQ with dedicated Satellite Partners.*

## HQ & Satellite Roles

- **OpenAA HQ (Megatrade):** Focuses on core software development, community management, PR, and high-level technical support.
- **Satellite Companies:** Focus on local sales, direct customer support, and the physical development of inspection equipment using the OpenAA brand and software.

This ecosystem allows satellite companies to bypass the massive R&D costs of building proprietary software, enabling them to generate high profit margins by focusing purely on solving their local customers' needs.

## Future Vision & Advanced Support



*Comprehensive risk mitigation, financial backing, and resource sharing.*

As the community grows, OpenAA HQ plans to provide robust backing for its satellites:

- **Financial Support:** Offering loans secured by sales receivables and credit guarantees to help satellites manage large equipment purchases and long collection periods.
- **Risk Mitigation:** Developing insurance mechanisms to handle the financial risks of advanced equipment returns.
- **Resource Sharing:** Creating a fluid system to share human resources and support staff among satellite companies to keep fixed costs low while maintaining high service quality.